

BME170

Lab-A: Molecular Testing Lab

PCR testing for the presence of fecal
contamination of water

Assigned Day: Week 7 – Thursday

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Lab Date: Thursdays, 5/15/25 & 5/22/25

I affirm, under penalty of law, that I completed the entire lab report on my own

by following the UCI Academic Integrity Policy at

<https://aisc.uci.edu/students/academic-integrity/index.php>.

Hypothesis and Purpose

The hypothesis of this experiment is that my unknown water sample contains bacterial DNA from at least one of the following microorganisms: *Escherichia coli*, *Serratia marcescens*, or *Bacillus subtilis*. These bacteria are often found in fecally contaminated or environmentally polluted water sources. The purpose of the experiment is to determine the presence of microbial contamination in my selected water sample through multiplex polymerase chain reaction (PCR), a technique that allows for the simultaneous amplification of DNA from multiple bacterial species. PCR results will be verified using agarose gel electrophoresis to visualize specific DNA bands corresponding to each microorganism.

Methods

Methods followed Edvotek protocol 'Water Quality Testing III: Multiplex PCR Testing of Water Contaminants' (Edvotek #953, Washington, D.C.). Four water samples were prepared: known samples of *E. coli*, *S. marcescens*, and *B. subtilis*, along with one unknown sample. Mine is a water sample taken from the sink of room MSTB 214 on 5/14/25 at 6:45 PM. Bacterial DNA was first isolated from each using a lysis buffer and heat treatment. After isolation, 5 µL of each DNA sample was mixed with 20 µL of a multiplex LyphoPrimer™ mix (containing primers specific to each bacterial species) and a PCR EdvoBead™ in a 0.2 mL snap-cap PCR tube. A positive control tube, containing all necessary components for a successful PCR, was also included.

The PCR protocol was as follows: initial denaturation at 94°C for 5 minutes; 30 cycles of denaturation at 94°C for 30 seconds, annealing at 54°C for 30 seconds, and extension at 72°C for 60 seconds; followed by a final extension at 72°C for 4 minutes. After PCR, the amplified DNA (amplicons) was separated using agarose gel electrophoresis and stained with SYBR® Safe. DNA bands were visualized under a blue-light transilluminator to identify the presence and identity of bacterial DNA based on band size.

Results

As shown in Figure 1, the positive control sample produced a cycle threshold (Cq) value of 22.93, indicating that amplification occurred, though the DNA concentration was relatively low. The fluorescence signal (RFU) was minimal, suggesting suboptimal amplification. This may have been caused by an insufficient amount of template DNA, inadequate mixing of reagents, or limited incorporation of SYBR™ Green. Additionally, potential inhibitors in the positive control preparation, such as leftover components from the DNA extraction process, could have interfered with the reaction's efficiency.

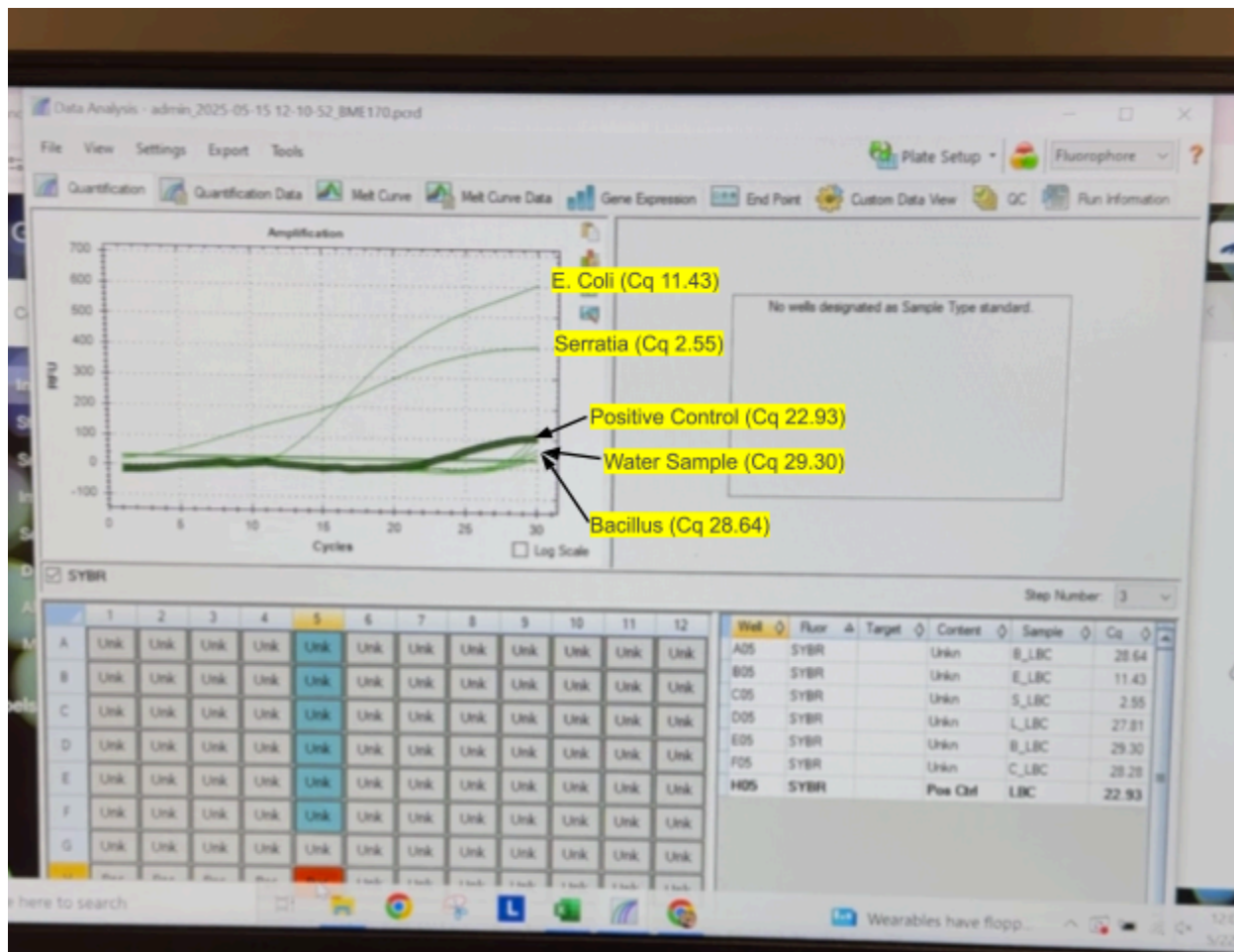


Figure 1: Amplification Data

For the known bacterial samples, amplification was observed across all wells. Notably, *Serratia marcescens* produced a very low Cq of 2.55, indicating a high starting concentration of DNA. *E. coli* also amplified efficiently, with a Cq of 11.43, while *Bacillus subtilis* and the positive control yielded Cq values near 28, reflecting lower template quantities.

My unknown sample taken from the lab sink, labeled as “Water Sample,” showed a Cq of 29.30, confirming the presence of amplifiable bacterial DNA, though at a lower abundance compared to

the other known samples. The relatively high C_q suggests the DNA was present but in smaller amounts, warranting further analysis to determine the bacterial identity.

Melt curve analysis, shown in Figure 2, was used to confirm amplification specificity. Each DNA product has a unique melting temperature (T_m), allowing comparison to known standards. A distinct melt peak at 83.5°C was detected in well B05 (E_LBC), consistent with *E. coli* DNA. This melting temperature strongly suggests that *E. coli* was successfully amplified from the water sample. The presence of *E. coli*, a common fecal indicator, supports the hypothesis that the water sample contains microbial contamination.

Other wells, including the positive control (H05), exhibited amplification but lacked clear melt peaks. This lack of specificity could be due to degraded DNA, non-specific amplification, or insufficient primer binding. The positive control's absence of a defined melt peak despite a measurable C_q value indicates potential issues with reaction composition or sample integrity.

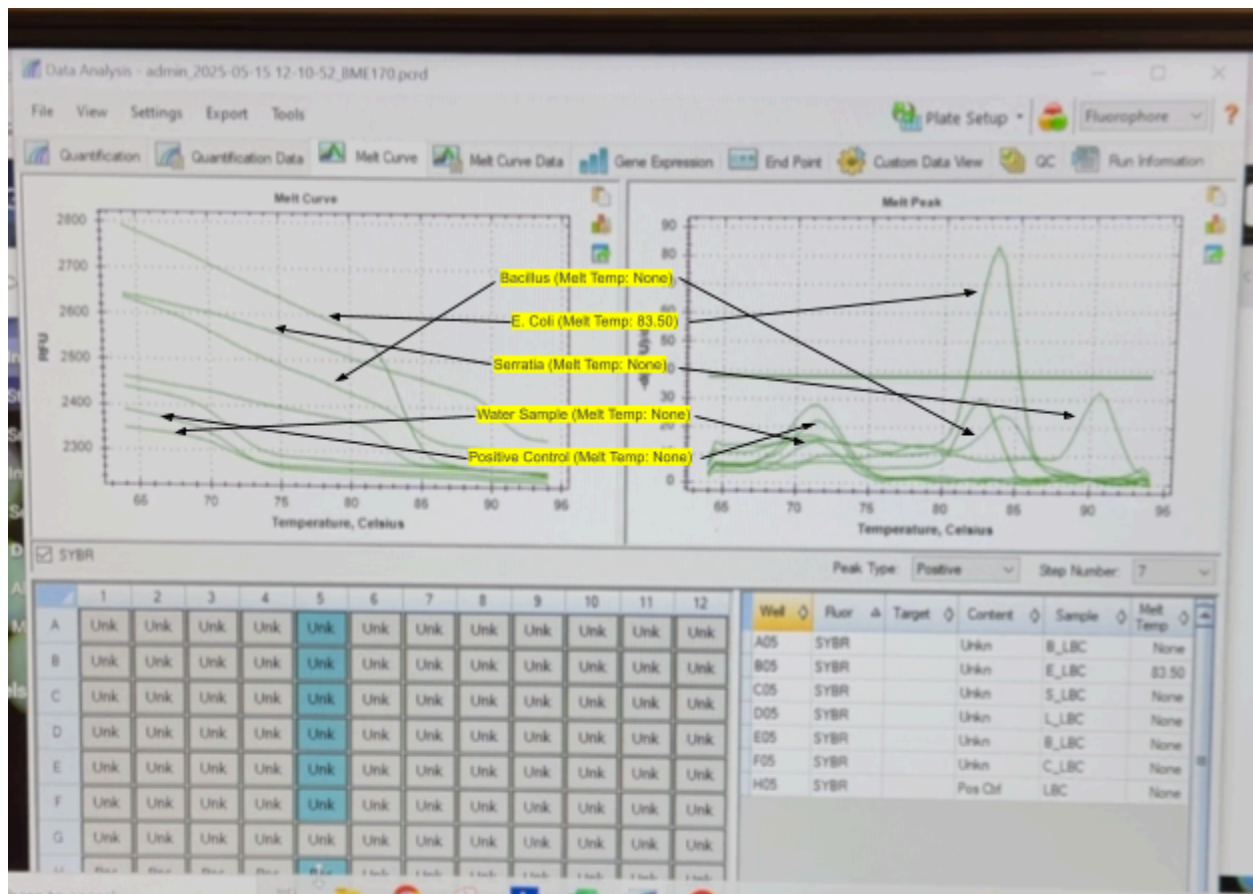


Figure 2: Melt Curve and Peak Data

Results from agarose gel electrophoresis (Figure 3) further validated PCR outcomes. In this analysis, DNA bands were visualized to compare the size and intensity of amplified products.

Samples producing concentrated, same-sized amplicons appear as bright, well-defined bands. Most unknown water samples displayed visible bands, suggesting successful amplification. However, my specific water sample (Lane 5) had a faint band with limited brightness, suggesting a lower concentration of DNA or non-specific amplification. No clear alignment with the *Serratia* or *Bacillus* lanes was observed.

Interestingly, the band in Lane 5 appeared to align closely with the *E. coli* band in Lane 2, reinforcing the conclusion drawn from qPCR and melt curve analysis. In contrast, the positive control and *Bacillus subtilis* samples did not produce visible bands, which may indicate amplification failure, insufficient DNA concentration, or problems with primer-template compatibility.

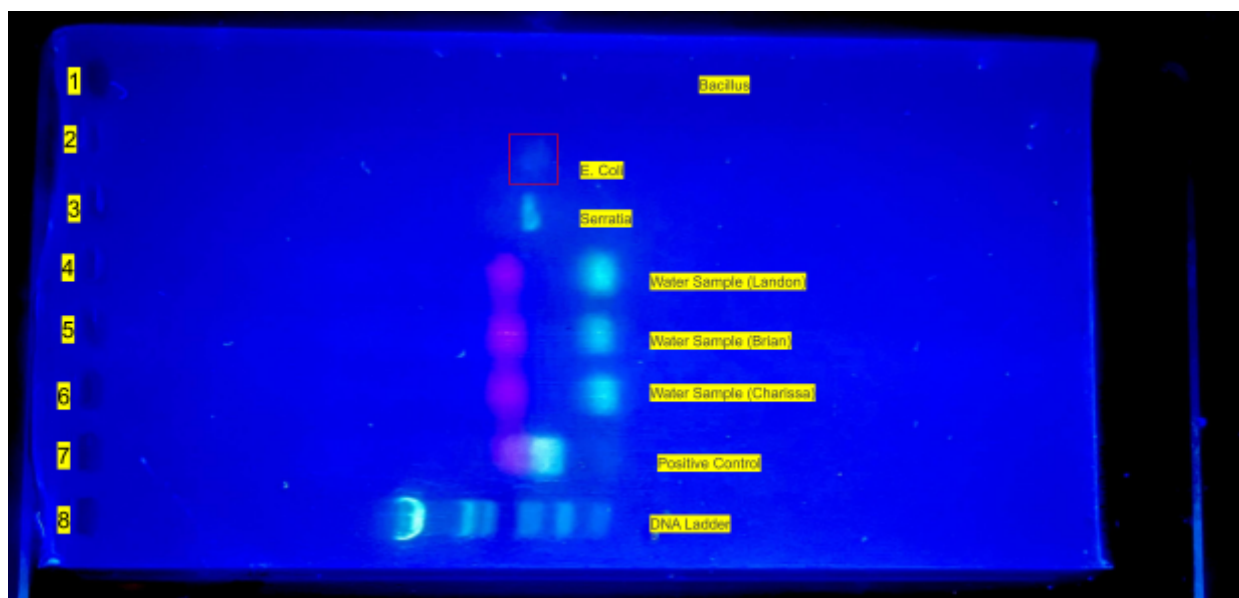


Figure 3: Results of Agarose Gel Electrophoresis

Overall, the gel electrophoresis confirmed successful amplification in multiple samples. The smallest DNA fragment appeared in the *Serratia* lane (Lane 3), as indicated by its greater migration distance. Bands from Charissa's and Landon's water samples migrated less, suggesting larger amplicons. The band in my water sample matched the position of *E. coli*, supporting previous evidence of its presence. The absence of bands in other samples underscores the importance of consistent technique and highlights possible limitations in reaction setup or reagent stability.

Conclusions and Discussion

The combined results from qPCR, melt curve analysis, and agarose gel electrophoresis provide partial support for the hypothesis that the unknown water sample contained bacterial DNA from at least one of the tested microorganisms. Specifically, strong evidence suggests the presence of

Escherichia coli in the water sample, indicated by a Cq value of 11.43, a defined melt temperature of 83.5°C, and a gel band that aligns with the known *E. coli* reference lane. These findings suggest that the water sample was contaminated with *E. coli*, a key indicator of fecal contamination, thereby supporting the hypothesis of microbial presence in the sample.

However, some results were inconclusive. The positive control produced a Cq value of 22.93, indicating some amplification, yet lacked a clear melt peak and did not produce a visible band in the gel. This discrepancy suggests potential issues with primer-template compatibility, insufficient mixing, or DNA degradation. Similarly, other bacterial controls, such as *Bacillus subtilis* and *Serratia marcescens*, did not show clearly interpretable results across all tests, possibly due to low starting template, reagent limitations, or technical errors during setup.

Interestingly, while the gel image showed a visible band in Lane 5 (water sample (Brian)), the band was not particularly bright and did not strongly match the known reference bands beyond the alignment with *E. coli*. This may point to either a lower concentration of bacterial DNA or possible non-specific amplification products. Melt curves for other samples also failed to produce clear peaks, reinforcing the possibility of non-specific amplification or inefficient primer binding.

To strengthen the conclusions of this experiment, further testing is recommended. Including additional reference samples of other common waterborne bacteria would improve the specificity of detection and help identify contaminants not captured by the current primer set. Repeating the PCR with fresh reagents and verifying DNA concentrations before amplification may also improve the clarity of results. Overall, while evidence of *E. coli* contamination was clearly demonstrated, further refinement of the experimental process and expanded testing could help paint a more complete picture of the microbial content in the water sample.